

Since $\{g_1, g_2\} = -\{g_2, g_1\}$, we have

$$\{\tilde{H}_h, H\} = 0.$$

This would imply that $\tilde{H}_h$ is actually, itself, a first integral of the molecular system. This certainly seems unlikely to hold except in very special cases indeed, unless the numerical method happens to coincide with the exact solution (up to a time rescaling). Thus the properties of symplecticness and energy conservation for numerical methods are essentially mutually exclusive from a practical point of view. A more precise formulation of this result was first given by Ge and Marsden [400].

Example 3.2 (7-Atom Lennard-Jones System) To test some of the methods presented in this chapter, let us see how they fare when we apply them to a system of $N = 7$ unit-mass particles moving in a pairwise potential $\varphi(r) = r^{-12} - 2r^{-6}$. This modified Lennard-Jones potential has stronger attraction (due to the factor of 2 multiplying r^{-6}) which tends to keep the system bound and makes testing a little easier (without worrying about boundary conditions or particle ejections). The Hamiltonian for our model is

$$H = \frac{1}{2} \sum_{i=1}^N \|\mathbf{p}_i\|^2 + U(\mathbf{q}), \quad U(\mathbf{q}) = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \varphi(\|\mathbf{q}_i - \mathbf{q}_j\|).$$

In our simulations we fixed the total momentum to be zero, so, although there may be net rotation, there is no translation of the atoms. It is easy to understand the minimum energy states for this system. At the global minimum potential energy configuration $U = E_0 = -12.5292$, the atoms are clustered in the compact arrangement with one at the center and the other six at the vertices of a regular hexagon. The positions of the outer atoms may be rotated without affecting the energy. At low total energy $E \approx E_0$, meaning that the system is started near the minimizing configuration, the particles remain in the hexagonal state, up to rotation and small vibrations. In order for two of the atoms to switch locations, it is necessary for the radial symmetry to be broken which requires crossing an energy “barrier.” For moderate energies, the atoms remain near the conformational states (local minima of potential) shown in Fig. 3.4 which are likewise separated by energy barriers.

A natural goal of simulation would be the computation of the relative probabilities of these various states. A more elementary task is to compute the radial distribution which gives the distribution of distance between atom pairs observed. The radial density function may be approximated from a histogram of all pair distances observed in a long simulation. (There are 21 at each step, so the amount of data is helpfully increased, reducing the “sampling error.”) This distribution is displayed in Fig. 3.5. The peaks of the radial distribution function are correlated with the various interatomic distances that appear in the cluster configurations shown in Fig. 3.4.